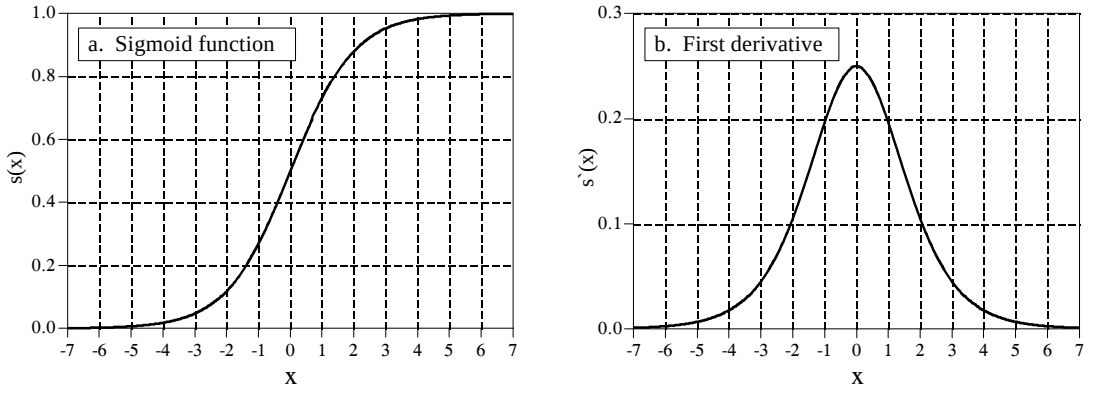

FIGURE 26-7
The sigmoid function and its derivative. Equations 26-1 and 26-2 generate these curves.

Figure 26-7a shows a closer look at the sigmoid function, mathematically described by the equation:

EQUATION 26-1
The sigmoid function. This is used in neural networks as a smooth threshold. This function is graphed in Fig. 26-7a.

$$s(x) = \frac{1}{1+e^{-x}}$$

The exact shape of the sigmoid is not important, only that it is a **smooth threshold**. For comparison, a **simple threshold** produces a value of *one* when $x > 0$, and a value of *zero* when $x < 0$. The sigmoid performs this same basic thresholding function, but is also *differentiable*, as shown in Fig. 26-7b. While the derivative is not used in the flow diagram (Fig. 25-5), it is a critical part of finding the proper weights to use. More about this shortly. An advantage of the sigmoid is that there is a shortcut to calculating the value of its derivative:

EQUATION 26-2
First derivative of the sigmoid function. This is calculated by using the value of the sigmoid function itself.

$$s'(x) = s(x)\,[\,1 - s(x)\,]$$

For example, if $x = 0$, then $s(x) = 0.5$ (by Eq. 26-1), and the first derivative is calculated: $s'(x) = 0.5(1 - 0.5) = 0.25$. This isn't a critical concept, just a trick to make the algebra shorter.

Wouldn't the neural network be more flexible if the sigmoid could be adjusted left-or-right, making it centered on some other value than $x = 0$? The answer is yes, and most neural networks allow for this. It is very simple to implement; an additional node is added to the input layer, with its input always having a